

Bingsheng Arthur Yao

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RESEARCH INTERESTS

My research lies at the intersection of **Human-Computer Interaction (HCI)** and **Natural Language Processing (NLP)**. I study how LLM agents can genuinely collaborate with humans through the lens of remote human collaboration, with a focus on designing the interactions and methodologies for effective human-agent collaboration in practice, and on developing and evaluating collaborative LLM agents.

APPOINTMENTS

2025 - Present	Associate Research Scientist Northeastern University
2024 - 2025	Postdoctoral Research Associate Northeastern University
2019 - 2024	Research Assistant Rensselaer Polytechnic Institute
2022 - 2022	Summer Research Intern IBM Research

EDUCATION

2019 - 2024	Rensselaer Polytechnic Institute Ph.D. in Computer Science Thesis: <i>Enhance Machine Reasoning via Active Learning with Human Rationales</i> Doctoral Committee: James Hendler (advisor), Dakuo Wang, Mohammed Zaki, Oshani Seneviratne
2018 - 2019	Rensselaer Polytechnic Institute M.S. in Information Technology
2014 - 2018	Rensselaer Polytechnic Institute B.S. in Computer Systems Engineering

GRANTS

Awarded	NIH R01 (R01AG097565) “BEACON: Bilingual AI to Enhance Caregivers’ Comprehension of Older Patients’ Clinical Course in Critical Care.”, (My Role: Multi-PI , PI: Prof. Yanjun Gao from University of Colorado Denver, Total: \$4,000,000)
Awarded	NIH R01 (R01CA301579) “Treatment-Related Cardiotoxicity Risk Prediction for Breast Cancer Patients with a Multimodal AI-based System”, (My Role: Co-I , PI: Prof. Weidan Cao from Ohio State University, Total: \$4,000,000)
Awarded	Microsoft Research Agentic AI Research and Innovation (AARI) “From Task Solvers to Teammates: A Theory-Grounded Architecture for Advancing Collaboration Readiness in LLM Agents”, (My Role: PI , Total: \$120,000 Azure Credit and \$40,000 Gift)

PUBLICATIONS

Conference

– 2026 –

- [C.32] Ziqi Yang, Yuxuan Lu, Jennifer Bagdasarian, Vedant Das Swain, Ritu Agarwal, Collin Campbell, Waddah Al-Refaie, Jehan El-Bayoumi, Guodong Gao, Dakuo Wang, **BINGSHENG YAO**, Shara Nawar (2026). “RECOVER: Designing a Large Language Model-based Remote Patient Monitoring System for Postoperative Gastrointestinal Cancer Care”. In: *Proceedings of the ACM on Human-Computer Interaction*
- [C.31] Mengxu Pan, Panxin Liu, Jinda Zhang, Raina Cao, Viduni Ariyawansa, Yaning Li, **BINGSHENG YAO**, Dakuo Wang, Philippe Pasquier, Alexandra Kitson (2026). “LLM-based Embodied Conversational Agent for Reducing Foreign Language Speaking Anxiety in Social VR”. in: *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems*
- [C.30] Menglin Zhao, Zhuorui Yong, Ruijia Guan, Kai-Wei Chang, Adrian Haimovich, Kei Ouchi, Timothy Bickmore, Zhan Zhang, **BINGSHENG YAO**, Dakuo Wang (2026). “Balancing Efficiency and Empathy: Healthcare Providers’ Perspectives on AI-Supported Workflows for Serious Illness Conversations in the Emergency Department”. In: *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems*
- [C.29] **BINGSHENG YAO**, Jiaju Chen, Chaoran Chen, April Yi Wang, Toby Jia-Jun Li, Dakuo Wang (2026). “Through the Lens of Human-Human Collaboration: An Configurable Research Platform for Exploring Human-Agent Collaboration”. In: *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems*, pp. 1–30
- [C.28] **BINGSHENG YAO**, Menglin Zhao, Zhan Zhang, Pengqi Wang, Emma G Chester, Changchang Yin, Tianshi Li, Varun Mishra, Lace M. Padilla, Odysseas P Chatzipanagiotou (2026). “Exploring Collaboration Breakdowns Between Provider Teams and Patients in Post-Surgery Care”. In: *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems*, pp. 1–17
- [C.27] Jingyu Tang, Chaoran Chen, Jiawen Li, Zhiping Zhang, Bingcan Guo, Ibrahim Khalilov, Simret Araya Gebreegziabher, **BINGSHENG YAO**, Dakuo Wang, Yanfang Ye (2026). “Dark patterns meet gui agents: Llm agent susceptibility to manipulative interfaces and the role of human oversight”. In: *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems*, pp. 1–26
- [C.26] Yuxuan Lu, Ting-Yao Hsu, Hansu Gu, Limeng Cui, Yaochen Xie, William P Headden III, **BINGSHENG YAO**, Akash Veeragouni, Jiapeng Liu, Sreyashi Nag (2026). “Agent A/B: Automated and Scalable A/B Testing on Live Websites with Interactive LLM Agents”. In: *Proceedings of the Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems*, pp. 1–12
- 2025 –
- [C.25] Maya Kruse, Shiyue Hu, Nicholas Derby, Yifu Wu, Samantha Stonbraker, **BINGSHENG YAO**, Dakuo Wang, Elizabeth Goldberg, Yanjun Gao (2025). “Large language models with temporal reasoning for longitudinal clinical summarization and prediction”. In: *Findings of ACL. EMNLP. Conference on Empirical Methods in Natural Language Processing*. Vol. 2025, pp. 20715–20735
- [C.24] Sze-yi Chan, Jiachen Li, **BINGSHENG YAO**, Amama Mahmood, Chien-Ming Huang, Holly Jimison, Elizabeth D Mynatt, Dakuo Wang (2025). “‘Mango Mango, How to Let The Lettuce Dry Without A Spinner?’: Exploring User Perceptions of Using An LLM-Based Conversational Assistant Toward Cooking Partner”. In: *Proceedings of the ACM on Human-Computer Interaction* 9.7, pp. 1–35
- [C.23] Jiachen Li, Xiwen Li, Justin Steinberg, Akshat Choube, **BINGSHENG YAO**, Xuhai Xu, Dakuo Wang, Elizabeth Mynatt, Varun Mishra (2025). “Vital Insight: Assisting Experts’ Context-Driven Sensemaking of Multi-modal Personal Tracking Data Using Visualization and Human-in-the-Loop LLM”. in: *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT)* ’25
- [C.22] Changchang Yin, Shihan Fu, **BINGSHENG YAO**, Thai-Hoang Pham, Weidan Cao, Dakuo Wang, Jeffrey Caterino, Ping Zhang (2025). “SepsisCalc: Integrating clinical calculators into early sepsis prediction via dynamic temporal graph construction”. In: *Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining V. 1*, pp. 2779–2790

- [C.21] Chaoran Chen, **BINGSHENG YAO**, Ruishi Zou, Wenyue Hua, Weimin Lyu, Toby Jia-Jun Li, Dakuo Wang (2025). “Towards a design guideline for rpa evaluation: A survey of large language model-based role-playing agents”. In: *Findings of the Association for Computational Linguistics: ACL 2025*
- [C.20] Zhiping Zhang, Chenxinran Shen, **BINGSHENG YAO**, Dakuo Wang, Tianshi Li (2025). “Secret use of large language model (LLM)”. in: *Proceedings of the ACM on Human-Computer Interaction (CSCW ’25)*
- [C.19] Hitesh Goel, Yoobin Park, Jin Liou, Darwin A Guevarra, Peggy Callahan, Jolene Smith, **BINGSHENG YAO**, Dakuo Wang, Xin Liu, Daniel McDuff (2025). “Promoting Prosociality via Micro-acts of Joy: A Large-Scale Well-Being Intervention Study”. In: *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI ’25)*
- [C.18] Yuling Sun, Jiaju Chen, Xiaomu Zhou, Xiaojuan Ma, **BINGSHENG YAO**, Kai Zhang, Liang He, Dakuo Wang (2025). “Live-Streaming-Based Dual-Teacher Classes for Equitable Education: Insights and Challenges From Local Teachers’ Perspective in Disadvantaged Areas”. In: *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI ’25)*
- [C.17] Jiaju Chen, Minglong Tang, Yuxuan Lu, **BINGSHENG YAO**, Elissa Fan, Xiaojuan Ma, Ying Xu, Dakuo Wang, Yuling Sun, Liang He (2025). “Characterizing LLM-Empowered Personalized Story Reading and Interaction for Children: Insights From Multi-Stakeholder Perspectives”. In: *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI ’25)*
- [C.16] Rudaiba Adnin, Atharva Pandkar, **BINGSHENG YAO**, Dakuo Wang, Maitraye Das (2025). “Examining Student and Teacher Perspectives on Undisclosed Use of Generative AI in Academic Work”. In: *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI ’25)*
- [C.15] Siyi Wu, Weidan Cao, Shihan Fu, **BINGSHENG YAO**, Ziqi Yang, Changchang Yin, Varun Mishra, Daniel Addison, Ping Zhang, Dakuo Wang (2025). “CardioAI: A Multi-modal AI-based System to Support Symptom Monitoring and Risk Detection of Cancer Treatment-Induced Cardiotoxicity”. In: *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI ’25)*
- 2024 –
- [C.14] Changchang Yin, Pin-Yu Chen, **BINGSHENG YAO**, Dakuo Wang, Jeffrey Caterino, Ping Zhang (2024). “Sepsislab: Early sepsis prediction with uncertainty quantification and active sensing”. In: *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD ’24)*
- [C.13] **BINGSHENG YAO**, Guiming Chen, Ruishi Zou, Yuxuan Lu, Jiachen Li, Shao Zhang, Yisi Sang, Sijia Liu, James Hendler, Dakuo Wang (2024). “More samples or more prompts? exploring effective few-shot in-context learning for LLMs with in-context sampling”. In: *Findings of the Association for Computational Linguistics: NAACL 2024 (NAACL ’24)*
- [C.12] Zhiping Zhang, Michelle Jia, Hao-Ping Lee, **BINGSHENG YAO**, Sauvik Das, Ada Lerner, Dakuo Wang, Tianshi Li (2024). “‘It’s a Fair Game’, or Is It? Examining How Users Navigate Disclosure Risks and Benefits When Using LLM-Based Conversational Agents”. In: *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems (CHI ’24)*
- [C.11] Shao Zhang, Jianing Yu, Xuhai Xu, Changchang Yin, Yuxuan Lu, **BINGSHENG YAO**, Melanie Tory, Lace M Padilla, Jeffrey Caterino, Ping Zhang (2024). “Rethinking human-AI collaboration in complex medical decision making: a case study in sepsis diagnosis”. In: *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems (CHI ’24)*
- [C.10] Ziqi Yang, Xuhai Xu, **BINGSHENG YAO**, Ethan Rogers, Shao Zhang, Stephen Intille, Nawar Shara, Guodong Gordon Gao, Dakuo Wang (2024). “Talk2care: An llm-based voice assistant for communication between healthcare providers and older adults”. In: *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT ’24)*
- [C.9] Xuhai Xu, **BINGSHENG YAO**, Yuanzhe Dong, Saadia Gabriel, Hong Yu, James Hendler, Marzyeh Ghassemi, Anind K Dey, Dakuo Wang (2024). “Mental-llm: Leveraging large language models for mental health prediction via online text data”. In: *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT ’24)*

- [C.8] Jiaju Chen, Yuxuan Lu, Shao Zhang, **BINGSHENG YAO**, Yuanzhe Dong, Ying Xu, Yunyao Li, Qianwen Wang, Dakuo Wang, Yuling Sun (2024). “StorySparkQA: Expert-Annotated QA Pairs with Real-World Knowledge for Children’s Story-Based Learning”. In: *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing (EMNLP ’24)*
- 2023 –
- [C.7] Ronald Cardenas, **BINGSHENG YAO**, Dakuo Wang, Yufang Hou (2023). “Don’t Get Too Technical with Me’: A Discourse Structure-Based Framework for Automatic Science Journalism”. In: *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP ’23)*
- [C.6] **BINGSHENG YAO**, Ishan Jindal, Lucian Popa, Yannis Katsis, Sayan Ghosh, Lihong He, Yuxuan Lu, Shashank Srivastava, Yunyao Li, James Hendler (2023). “Beyond Labels: Empowering Human Annotators with Natural Language Explanations through a Novel Active-Learning Architecture”. In: *Findings of the Association for Computational Linguistics: EMNLP 2023 (EMNLP ’23)*
- [C.5] **BINGSHENG YAO**, Prithviraj Sen, Lucian Popa, James Hendler, Dakuo Wang (2023). “Are Human Explanations Always Helpful? Towards Objective Evaluation of Human Natural Language Explanations”. In: *Proceedings of the 61th Annual Meeting of the Association for Computational Linguistics (ACL ’23)*
- 2022 AND BEFORE –
- [C.4] Zheng Zhang, Ying Xu, Yanhao Wang, **BINGSHENG YAO**, Daniel Ritchie, Tongshuang Wu, Mo Yu, Dakuo Wang, Toby Jia-Jun Li (2022). “Storybuddy: A human-ai collaborative chatbot for parent-child interactive storytelling with flexible parental involvement”. In: *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems (CHI ’22)*
- [C.3] Ying Xu*, Dakuo Wang*, Mo Yu*, Daniel Ritchie*, **BINGSHENG YAO***, Tongshuang Wu, Zheng Zhang, Toby Jia-Jun Li, Nora Bradford, Branda Sun (2022). “Fantastic Questions and Where to Find Them: FairytaleQA–An Authentic Dataset for Narrative Comprehension”. In: *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (ACL ’22)*
- [C.2] **BINGSHENG YAO***, Dakuo Wang*, Tongshuang Wu, Tran Hoang, Branda Sun, Toby Jia-Jun Li, Mo Yu, Ying Xu (2022). “It is ai’s turn to ask human a question: Question and answer pair generation for children storybooks in fairytaleqa dataset”. In: *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (ACL ’22)*
- [C.1] Jaimie Drozdal, Justin Weisz, Dakuo Wang, Gaurav Dass, **BINGSHENG YAO**, Changruo Zhao, Michael Muller, Lin Ju, Hui Su (2020). “Trust in AutoML: exploring information needs for establishing trust in automated machine learning systems”. In: *Proceedings of the 25th international conference on intelligent user interfaces (IUI ’20)*
- Journals**
- [J.5] Jiahao Xu, Changchang Yin, Odysseas P Chatzipanagiotou, Diamantis I Tsilimigras, Kevin Clear, **BINGSHENG YAO**, Weidan Cao, Dakuo Wang, Timothy M Pawlik, Ping Zhang (2026). “SurgWound-Bench: a benchmark for surgical wound diagnosis”. In: *npj Digital Medicine*
- J.4 Chaoran Chen, Zhiping Zhang, Bingcan Guo, Shang Ma, Ibrahim Khalilov, Simret A Gebreegziabher, **BINGSHENG YAO**, Dakuo Wang, Yanfang Ye, Ziang Xiao (2026). “The Obvious Invisible Threat: Vulnerabilities of LLM-Powered GUI Agents to Adversarial UI Manipulations in Web Interaction Tasks”. In: *ACM Transactions on AI Security and Privacy*
- J.3 Jiessie Tie, **BINGSHENG YAO**, Tianshi Li, Hongbo Fang, Syed Ishtiaque Ahmed, Dakuo Wang, Shurui Zhou (2026). ““Should I Give Up Now?” Investigating LLM Pitfalls in Software Engineering”. In: *ACM Transactions on Software Engineering and Methodology*
- [J.2] Amama Mahmood, Junxiang Wang, **BINGSHENG YAO**, Dakuo Wang, Chien-Ming Huang (2025). “User interaction patterns and breakdowns in conversing with LLM-powered voice assistants”. In: *International Journal of Human-Computer Studies*

- [J.1] Xiangyang Mou*, Chenghao Yang*, Mo* Yu, **BINGSHENG YAO**, Xiaoxiao Guo, Saloni Potdar, Hui Su (2021). “Narrative question answering with cutting-edge open-domain qa techniques: A comprehensive study”. In: *Transactions of the Association for Computational Linguistics* 9, pp. 1032–1046
- Workshops**
- W.4 **BINGSHENG YAO**, Chaoran Chen, April Yi Wang, Tongshuang Wu, Toby Jia-Jun Li, Dakuo Wang (2026). “From Human-Human Collaboration to Human-Agent Collaboration: A Vision, Design Philosophy, and an Empirical Framework for Achieving Successful Partnerships Between Humans and LLM Agents”. In: *Proceedings of the Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems*, pp. 1–6
- [W.3] Hongyu Wan, Jinda Zhang, Abdulaziz Arif Suria, **BINGSHENG YAO**, Dakuo Wang, Yvonne Coady, Mirjana Prpa (2024). “Building llm-based ai agents in social virtual reality”. In: *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI ’24)*
- [W.2] Mirjana Prpa, Giovanni Troiano, **BINGSHENG YAO**, Toby Jia-Jun Li, Dakuo Wang, Hansu Gu (2024). “Challenges and Opportunities of LLM-Based Synthetic Personae and Data in HCI”. in: *Companion Publication of the 2024 Conference on Computer-Supported Cooperative Work and Social Computing (CSCW ’24 Workshop)*
- [W.1] Tianshi Li, Sauvik Das, Hao-Ping Lee, Dakuo Wang, **BINGSHENG YAO**, Zhiping Zhang (2024). “Human-centered privacy research in the age of large language models”. In: *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI ’24 Special Interest Groups)*

SERVICES

Area Chair

ACL Rolling Review (Starting Jun. 24)

ACM CHI (’25, ’26)

ACM CSCW (’25, ’26)

Conference Reviewer

ACL Rolling Review (from Aug 23 to Feb 24); EMNLP ’23; NAACL ’24; ACM IMWUT (’23, ’24); ACM CHI ’24; ACM IUI ’24

Journals Reviewer

Natural Human Behavior

International Journal of Human–Computer Interaction (IJHCI)

International Journal of Human-Computer Studies (IJHCS)

Information Process & Management

CCF Transactions on Pervasive Computing and Interaction (TPCI)

TEACHING

Graduate TA

2018 Fall **CSCI 2500 Computer Organization (201 Students)**
Rensselaer Polytechnic Institute

2019 Spring **CSCI 4150 Introduction to Artificial Intelligence (182 Students)**
Rensselaer Polytechnic Institute

2023 Fall **CSCI 2500 Computer Organization (344 Students)**
Rensselaer Polytechnic Institute

Guest Lectures

2024 Spring **Human-Centered AI for Sepsis Early Diagnosis**
University of Southern California

2024 Spring	Human-Centered NLP Systems Northeastern University
2024 Fall	Human-Centered NLP Systems Northeastern University

AWARDS & HONORS

2019 - Current	Member of Upsilon Pi Epsilon The International Honor Society for the Computing and Information Disciplines
2019 - Current	Member of Gamma Nu Eta The National Information Technology Honor Society
2022 - 2023	Artificial Intelligence Research Collaboration Fellowship IBM Research

MENTORSHIP

2023 - Current	Yuxuan Lu Ph.D. student at Northeastern
2023 - Current	Jiaju Chen Research Assistant at Northeastern (Currently Master's student at Rice)
2023 - Current	Ruishi Zou Undergraduate student at Tongji (Currently Master student at UCSD)
2023 - Current	Ziqi Yang Master student at UMich (Currently Ph.D. student at UC Irvine)
2024 - Current	Bo Sun Research Assistant at Northeastern
2024 - Current	Shihan Fu Ph.D. student at Northeastern
2024 - Current	Ziyi Wang Ph.D. student at Northeastern
2024 - Current	Menglin Zhao Ph.D. student at Northeastern
2024 - Current	Siyi Wu Ph.D. student at University of Toronto
2023 - 2024	Zhiping Zhang Research Assistant at Northeastern (Currently Ph.D. student at Northeastern)
2023 - 2024	Kimberly Do Ph.D. student at Northeastern
2023 - 2024	Jiayi Lu Master student at Northeastern